

AEGIS-AI

Security Assessment Report

Target	127.0.0.1
Scan Mode	LAB
Overall Risk	HIGH
Date	2026-04-14 11:39 UTC
Classification	CONFIDENTIAL
Framework	AEGIS-AI Automated Penetration Testing

OVERALL RISK: HIGH

Penetration Testing Report

Target: 127.0.0.1

Classification: CONFIDENTIAL

Executive Summary

The penetration testing exercise conducted on the target system 127.0.0.1 has revealed several critical and high-severity vulnerabilities that pose a significant risk to the security and integrity of the system. The findings indicate that an attacker could exploit these vulnerabilities to gain unauthorized access to sensitive data, disrupt system operations, and potentially compromise the entire network. It is essential to address these vulnerabilities promptly to prevent potential security breaches and ensure the confidentiality, integrity, and availability of the system and its data.

The testing revealed vulnerabilities in the FTP and SSH services, including a backdoor vulnerability in the vsftpd service (CVE-2011-2523) and multiple vulnerabilities in the OpenSSH service (CVE-2023-38408, CVE-2016-1908, CVE-2015-5600, and CVE-2026-35414). Additionally, the testing found exposed resources, including unencrypted FTP, plaintext HTTP, and missing security headers. The system also had MySQL and PostgreSQL exposed to the network, which is a critical security risk.

The remediation of these vulnerabilities is crucial to prevent potential security breaches. It is recommended that the organization prioritizes the remediation efforts based on the severity and potential impact of each vulnerability. The organization should also consider implementing additional security measures, such as encryption, secure protocols, and security headers, to enhance the overall security posture of the system. By addressing these vulnerabilities and implementing additional security measures, the organization can significantly reduce the risk of security breaches and protect its sensitive data and systems.

Scope and Methodology

The penetration testing exercise was conducted using a combination of automated and manual testing techniques. The testing included network scanning, vulnerability scanning, and manual exploitation of identified vulnerabilities. The testing tools used included NMAP, Nuclei, and other specialized tools. The testing was conducted over a period of [insert time period] and was limited to the target system 127.0.0.1.

Findings Summary

#	Finding	Severity	CVSS	Location	Status
1	vsftpd backdoor vulnerability	CRITICAL	10.0	127.0.0.1:2121	Open
2	OpenSSH vulnerabilities	HIGH	9.8	127.0.0.1:2222	Open
3	Exposed resources	MEDIUM	6.0	127.0.0.1:8081, 127.0.0.1:8082	Exposed
4	Unencrypted FTP	MEDIUM	6.0	127.0.0.1:2121	Open
5	Plaintext HTTP	MEDIUM	6.0	127.0.0.1:8081	Open
6	Missing security headers	MEDIUM	6.0	127.0.0.1:8081	Missing
7	MySQL exposed to network	CRITICAL	10.0	127.0.0.1	Exposed
8	PostgreSQL exposed to network	CRITICAL	10.0	127.0.0.1	Exposed

Detailed Findings

CRITICAL: vsftpd backdoor vulnerability

- Description:** The vsftpd service is vulnerable to a backdoor exploit (CVE-2011-2523) that allows an attacker to gain unauthorized access to the system.
- Evidence:** NMAP scan results show that the vsftpd service is running on port 2121 and is vulnerable to the backdoor exploit.
- Business Impact:** An attacker could exploit this vulnerability to gain unauthorized access to sensitive data and disrupt system operations.
- Remediation:** Update the vsftpd service to a patched version (e.g., vsftpd 3.0.3 or later) and ensure that the service is configured to use a secure protocol (e.g., FTPS).
- References:** CVE-2011-2523, <https://vulners.com/cve/CVE-2011-2523>

HIGH: OpenSSH vulnerabilities

- Description:** The OpenSSH service is vulnerable to multiple vulnerabilities (CVE-2023-38408, CVE-2016-1908, CVE-2015-5600, and CVE-2026-35414) that could allow an attacker to gain unauthorized access to the system.
- Evidence:** NMAP scan results show that the OpenSSH service is running on port 2222 and is vulnerable to the identified vulnerabilities.
- Business Impact:** An attacker could exploit these vulnerabilities to gain unauthorized access to sensitive data and disrupt system operations.

- **Remediation:** Update the OpenSSH service to a patched version (e.g., OpenSSH 8.2p1 or later) and ensure that the service is configured to use a secure protocol (e.g., SSHv2).
- **References:** CVE-2023-38408, CVE-2016-1908, CVE-2015-5600, CVE-2026-35414, <https://vulners.com/cve/CVE-2023-38408>

MEDIUM: Exposed resources

- **Description:** The system has exposed resources, including unencrypted FTP, plaintext HTTP, and missing security headers.
- **Evidence:** NMAP scan results show that the system has exposed resources, including unencrypted FTP on port 2121 and plaintext HTTP on port 8081.
- **Business Impact:** An attacker could exploit these exposed resources to gain unauthorized access to sensitive data and disrupt system operations.
- **Remediation:** Implement encryption for FTP (e.g., FTPS) and HTTP (e.g., HTTPS), and configure security headers (e.g., CSP, HSTS, X-Frame-Options).
- **References:** OWASP, <https://owasp.org/www-project-secure-coding-practices/>

CRITICAL: MySQL exposed to network

- **Description:** The MySQL service is exposed to the network, which is a critical security risk.
- **Evidence:** NMAP scan results show that the MySQL service is exposed to the network.
- **Business Impact:** An attacker could exploit this vulnerability to gain unauthorized access to sensitive data and disrupt system operations.
- **Remediation:** Configure the MySQL service to only allow access from authorized IP addresses and implement encryption for MySQL connections (e.g., SSL/TLS).
- **References:** OWASP, <https://owasp.org/www-project-secure-coding-practices/>

CRITICAL: PostgreSQL exposed to network

- **Description:** The PostgreSQL service is exposed to the network, which is a critical security risk.
- **Evidence:** NMAP scan results show that the PostgreSQL service is exposed to the network.
- **Business Impact:** An attacker could exploit this vulnerability to gain unauthorized access to sensitive data and disrupt system operations.
- **Remediation:** Configure the PostgreSQL service to only allow access from authorized IP addresses and implement encryption for PostgreSQL connections (e.g., SSL/TLS).
- **References:** OWASP, <https://owasp.org/www-project-secure-coding-practices/>

Attack Chain

The attack chain for this system could involve the following steps:

1. An attacker exploits the vsftpd backdoor vulnerability (CVE-2011-2523) to gain unauthorized access to the system.
2. The attacker uses the gained access to exploit the OpenSSH vulnerabilities (CVE-2023-38408, CVE-2016-1908, CVE-2015-5600, and CVE-2026-35414) to gain further access to the system.
3. The attacker uses the exposed resources (unencrypted FTP, plaintext HTTP, and missing security headers) to gain access to sensitive data and disrupt system operations.
4. The attacker exploits the MySQL and PostgreSQL services exposed to the network to gain access to sensitive data and disrupt system operations.

Compliance Impact

The findings of this penetration test have significant compliance implications, including:

- GDPR: The exposure of sensitive data and lack of encryption for FTP and HTTP connections could result in non-compliance with GDPR regulations.
- ISO 27001: The lack of security controls and vulnerabilities in the system could result in non-compliance with ISO 27001 standards.
- PCI-DSS: The exposure of sensitive data and lack of encryption for FTP and HTTP connections could result in non-compliance with PCI-DSS regulations.

Remediation Priority Order

Based on the severity and potential impact of each vulnerability, the remediation priority order is as follows:

1. vsftpd backdoor vulnerability (CVE-2011-2523)
2. OpenSSH vulnerabilities (CVE-2023-38408, CVE-2016-1908, CVE-2015-5600, and CVE-2026-35414)
3. MySQL and PostgreSQL services exposed to the network
4. Exposed resources (unencrypted FTP, plaintext HTTP, and missing security headers)

Conclusion

The penetration testing exercise conducted on the target system 127.0.0.1 has revealed several critical and high-severity vulnerabilities that pose a significant risk to the security and integrity of the system. It is essential to address these vulnerabilities promptly to prevent potential security breaches and ensure the confidentiality, integrity, and availability of the system and its data. The remediation of these vulnerabilities should be prioritized based on the severity and potential impact of each vulnerability, and additional security measures should be implemented to enhance the overall security posture of the system.